

Progressive block-size sweeps on TU-Berlin sketches

Five families $\times$ two inits $\times$ two compression rates ($m = n = 8, 256 \times 256$)

Generated 2026-05-25. Mean test PSNR (dB) over 50 sketches. — = still training; n/a = stage undefined (mera needs $k \in \{2, 4, 8\}$).

1 Sketches are a different regime

The TU-Berlin set is line drawings: $\approx 96.5\%$ white pixels with thin black ink. They are **block-sparse** — most 8×8 tiles are blank — so a classical block transform is near-lossless: block-DCT-8 reaches 100.65 dB at $\rho = 0.2$ (median 94.48), versus ≈ 32 dB on DIV2K. This inverts the DIV2K story, so we report the learned circuits at **two** compression rates: the headline $\rho = 0.20$ ($5\times$) and an aggressive $\rho = 0.05$ ($20\times$).

The experiment is otherwise identical to the DIV2K matrix: each stage k trains a $2^k \times 2^k$ inner circuit replicated by BlockedBasis (bare at $k = 8$), 1008 steps, MSELoss top- k . Inits: **identity** (gates dropped to identity operator) and **random** (rich Haar $U(2)/U(4)$; qft Haar-on-H + random phases; tebd/entangled_qft/mera native seed). Families run $k = 2..8$ except mera ($k \in \{2, 4, 8\}$).

2 Light compression: $\rho = 0.20$ ($5\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	81.1	78.8	74.9	62.1	62.1	61.5	61.5
rich	random	79.5	73.8	62.6	48.8	41.5	33.9	44.9
qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
qft	random	87.1	83.1	65.6	48.5	39.6	43.2	44.2
tebd	identity	85	87.5	69.6	51.4	50.2	62.9	62.9
tebd	random	88.2	87.2	69.5	45.9	34.6	44.2	38.1
ent-qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
ent-qft	random	82.7	85.4	70.1	45.7	38.4	44.2	44.2
mera	identity	85	n/a	69.6	n/a	n/a	n/a	62.9
mera	random	88.2	n/a	69.4	n/a	n/a	n/a	38.4

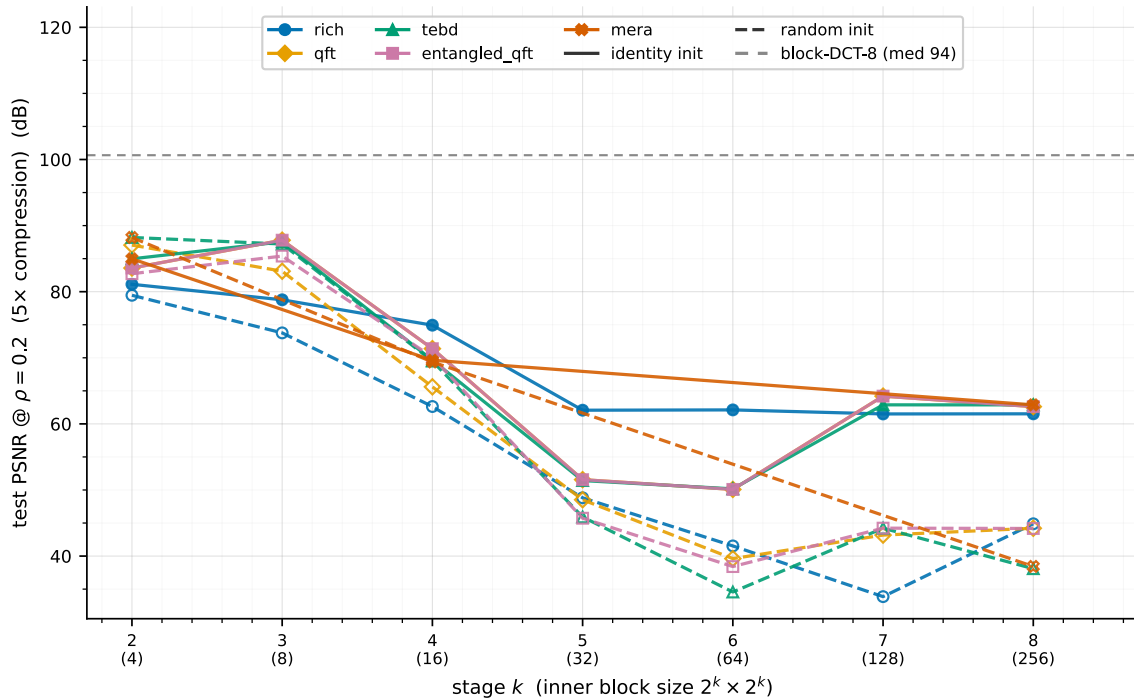


Figure 1: PSNR @ $\rho = 0.20$. Block-sparse sketches make **small** blocks win: the curve falls with k . All learned circuits sit below classical block-DCT-8 (grey).

3 Heavy compression: $\rho = 0.05$ (20 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	7.6	37.3	37.2	34.8	35.5	35.5	36.1
rich	random	7.6	30.6	31.6	29.6	28.5	25.7	29.3
qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
qft	random	7.6	29.8	31	28.8	27.3	28	28.4
tebd	identity	7.6	33.5	31.5	29.7	29.7	31.5	31.5
tebd	random	7.6	33.5	31.5	28.5	26.1	28.5	27.1
ent-qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
ent-qft	random	7.6	33.5	31.5	28.5	27.1	28.5	28.5
mera	identity	7.6	n/a	31.5	n/a	n/a	n/a	31.5
mera	random	7.6	n/a	31.5	n/a	n/a	n/a	27.1

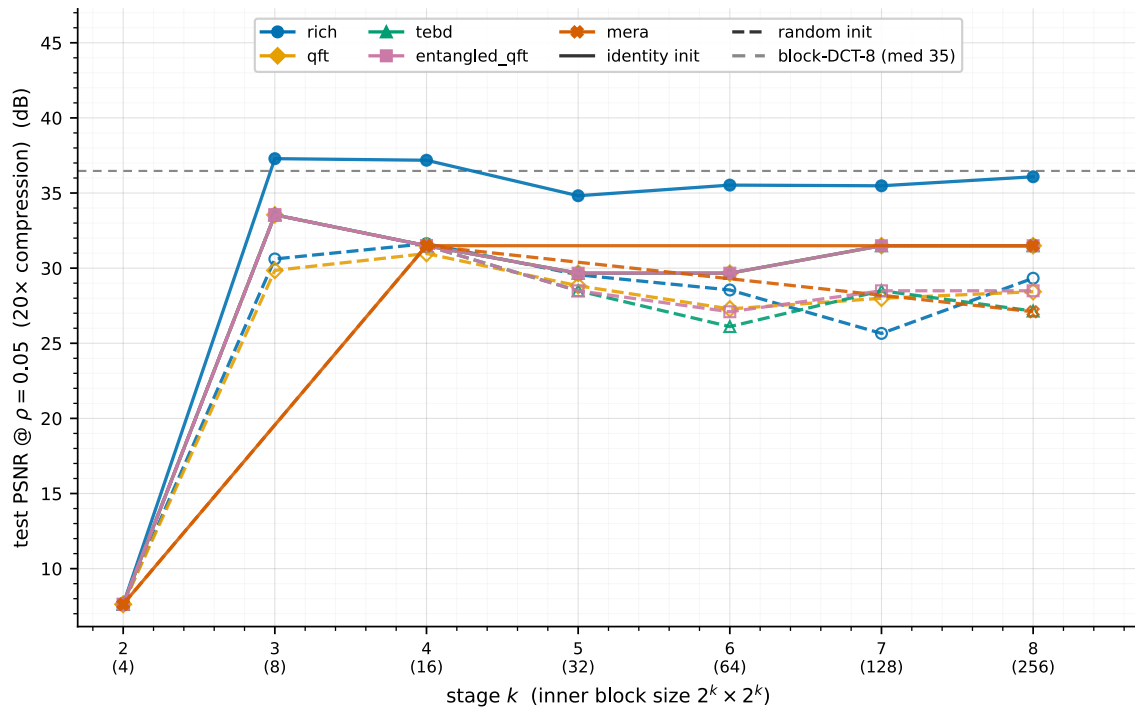


Figure 2: PSNR @ $\rho = 0.05$. At 20 $\times$ compression the learned circuits become **competitive with** block-DCT-8 (grey) — rich (identity) edges above it at $k = 3, 4$. The per- k curve also flattens.

4 Very heavy compression: $\rho = 0.01$ (100 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	0.9	4.8	4.8	22.1	21.7	22.2	22.9
rich	random	0.9	1.9	18.7	21.8	22.9	22.1	23
qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
qft	random	0.9	1.9	21.8	22.9	22.5	22.5	23.1
tebd	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
tebd	random	0.9	4.8	23.2	23.2	22.3	23.2	22.7
ent-qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
ent-qft	random	0.9	4.8	23.2	23.2	22.7	23.2	23.2
mera	identity	0.9	n/a	23.2	n/a	n/a	n/a	23.2
mera	random	0.9	n/a	23.2	n/a	n/a	n/a	22.7

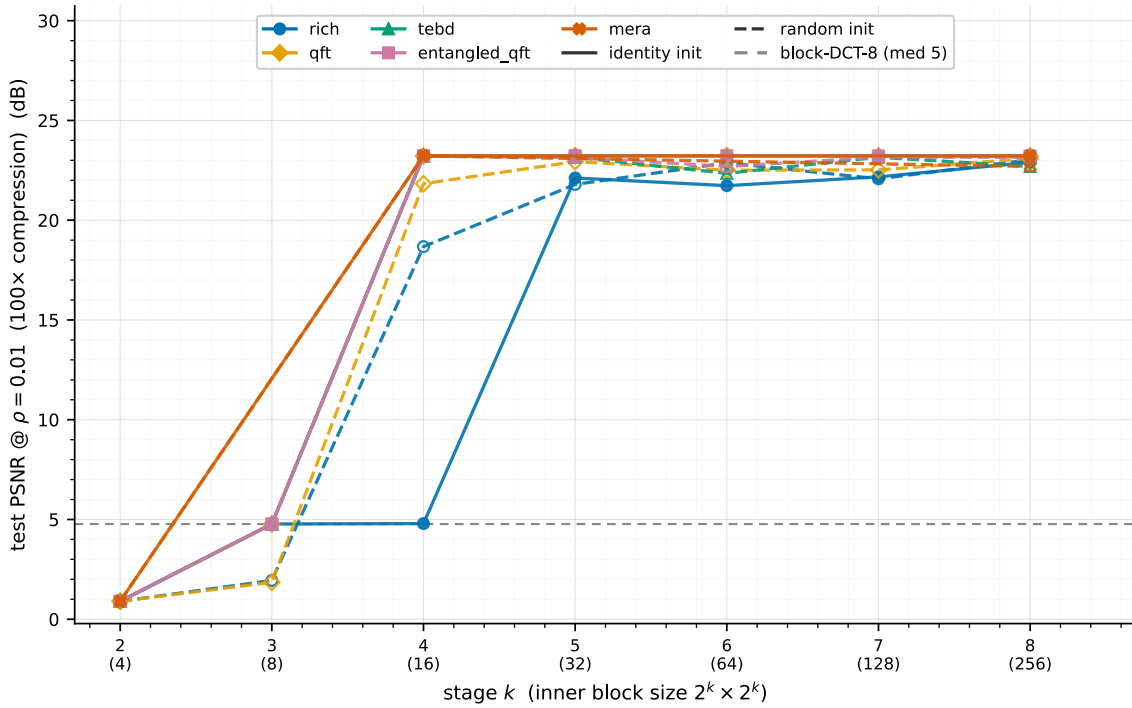


Figure 3: PSNR @ $\rho = 0.01$. block-DCT-8 **collapses** to ≈ 4.8 dB; the learned circuits plateau ≈ 23 dB, clearing it by ≈ 18 dB. The QFT-derived families edge rich here. $k = 2$ degenerates (too few coefficients survive $100 \times$ pooling on a 4×4 block).

5 Observations

- **Block sparsity dominates the light-compression regime, and the curve inverts.** At $\rho = 0.20$ PSNR **falls** with block size k — small blocks win because each blank tile costs almost no coefficients — the opposite of DIV2K’s flat ≈ 33 dB. The best learned circuits peak at ≈ 88 dB ($k = 3$, the QFT-derived families) but still trail classical block-DCT-8 (median ≈ 94 dB): on near-lossless-compressible sketches the fixed block transform is hard to beat.
- **The family ranking flips between rates.** At $\rho = 0.20$ the QFT-derived families (qft/tebd/entangled_qft) **beat** rich at the small blocks (≈ 88 vs 79 dB at $k = 3$). At $\rho = 0.05$ this reverses: rich (identity) leads at ≈ 37 dB and even edges **above** block-DCT-8 (≈ 36 dB) at $k = 3, 4$. When only 5% of coefficients survive, learning a genuine energy-compacting transform pays off and the classical near-lossless advantage evaporates.
- **Identity init beats random for every family, by a lot.** On sketches the gap is far larger than on DIV2K — up to ≈ 25 – 30 dB at large k ($\rho = 0.2$, e.g. rich $k = 7$: 61.5 vs 33.9). A random start is genuinely damaging here.

- **The QFT-family identity-init equivalence partly survives.** Under identity init qft and entangled_qft are bit-identical at every k (as on DIV2K); at $\rho = 0.05$ all four QFT-derived families coincide, but at $\rho = 0.20$ they split into a qft/entangled_qft pair and a tebd/mera pair. rich (complex $U(4)$) stands apart from all of them.

Working artefact (report-first; not yet cellified into results/). All combos from /tmp/tuberlin/<family>_<init>; classical refs computed on the same seed=42 / 50-sketch test split.